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TBT–401

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

BIOPHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the role of secondary bonds (weak interactions) in the DNA structure. (CO1)

(b) Discuss the hybridization in biomolecules with the help of an example. (CO1)

(c) Explain the Laue method in the X-ray diffraction technique. (CO1)

2. (a) What is meant by polarization of light. Explain its role in circular dichroism (CD) spectroscopy. (CO2)

(b) Draw the Ramachandran plot for alanine and glycine and discuss its rotational conformational in terms of φ and ψ. (CO2)

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(c) Explain the spontaneous and non-spontaneous process in terms of enthalpy, entropy, and Gibbs free energy. (CO2)

3. (a) Describe the structure of plasma membrane and mechanism of resting membrane potential in cell membrane. (CO3)

(b) Explain the Nernst equation ? Explain the Goldman-Hodgkin-Katz voltage equation. (CO3)

(c) Draw and describe the electrical equivalent circuit model for membrane potential. (CO3)

4. (a) Explain the current advances adopted for the treatment of neurological disorders. (CO4)

(b) "Materials of which a single unit is sized between 1 and 100 nm." Comment on the statement given and also highlight the key role of such materials in the field of biology. (CO4)

(c) Give an overview of the structure and movement of cilia and flagella. (CO4)

5. (a) What are molecular motors ? Diagrammatically describe the mechanism of its functioning with examples. (CO5)

(b) Discuss the term mechanobiology and its importance in human health. (CO5)

(c) Discuss components of vascular system and disease related to it. (CO5)

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TBT–402

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

IMMUNOLOGY AND IMMUNOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the alternative pathway of synthesis of complement proteins, and their mode of action against bacterial pathogen. (CO2)

(b) What are antibodies ? Describe affinity and avidity in terms of antigen and antibody interactions. (CO1)

(c) Using suitable diagrams, differentiate the MHCs based on their structure and antigen presentation. (CO2)

2. (a) What are DNA vaccines ? Using a suitable diagram, explain the process of developing these vaccines. (CO3)

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(b) Describe in detail types of innate immune response. Also comment on phagocytosis barrier of innate immune response. (CO1)

(c) Describe Complement fixation test and its applications in immunodiagnostics of an antigen. (CO4)

3. (a) What are subunit vaccines ? Describe the types of vaccines currently in use for immunization against COVID-19 infection ? (CO3)

(b) Explain the role of immune complexes in inducing Type-III hypersensitivity reaction in human body. (CO5)

(c) Explain the contribution of different scientists in the field of immunology. (CO1)

4. (a) Describe RIA and its significance in determination of Hepatitis B infection in humans. (CO4)

(b) What is Chemiluminescence ? Describe the technique, its principle and application in the identification of an antigen. (CO4)

(c) What are MALT ? Explain the role in providing defense against the pathogens. (CO1)

5. (a) What are auto-immune disorders ? Explain the cause of Hashimoto's disease and Type-I diabetes. (CO5)

(b) Explain the types of transplant antigens ? Describe the mechanism of transplant rejection by T-lymphocytes. (CO5)

(c) What is allergy ? Explain the mechanism of Type-I hypersentivity and its clinical features. (CO5)

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TBT–403

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

BIO-ANALYTICAL TECHNIQUE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain 10 industrial applications of bioanalytical techniques. (CO1)

   (b) Explain Atomic force microscopy. (CO1)

   (c) Explain marker enzymes. (CO1)

2. (a) Explain Isoelectric focusing in detail. (CO2)

   (b) Explain the working principle of ultracentrifugation and its application. (CO2)

   (c) Write short notes on the following : (CO2)

   (i) Gel Electrophoresis

   (ii) Paper Electrophoresis

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3. (a) Discuss the working principle of ion-exchange chromatography. (CO3)

   (b) Discuss TLC and its application in detail. (CO3)

   (c) Write a short note on gas chromatography and its application. (CO3)

4. (a) Write a short note on working principal of spectrophotometry. (CO4)

   (b) Discuss the working principle of Mass spectroscopy and its various applications. (CO4)

   (c) Write a short note on X-ray diffraction. (CO4)

5. (a) Enlist the properties of α, β and γ rays and their uses. (CO5)

   (b) Write a short note on radioactivity measurement method. (CO5)

   (c) Write a short note on in vivo and in vitro labelling technique. (CO5)

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TBT–404

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

ENVIRONMENTAL BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Highlight the role of degradation mechanism of xenobiotic compounds. (CO4, CO5)

(b) Diagrammatically explain the photocatalysis process mediated by Titanium dioxide particles. (CO4, CO5)

(c) Discuss the strategies to be follqwed for restoration of waste lands. (CO4, CO5)

2. (a) What are the most common prejudices against enzymes ? (CO3, CO5)

(b) "A tool used to assess the significant effects of a project or development proposal on the environment." Comment on the afore mentioned statement highlight its objectives. (CO3, CO5)

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(c) What are genetically modified organisms ? Explain the negative impacts imposed by GMOs on health and the environment. (CO3, CO5)

3. (a) Explain any *two* degradative mechanisms of plastics with example. (CO5, CO2)

(b) Differentiate between Rotating Biological Contractors and Trickling Filters. (CO5, CO2)

(c) Describe the construction design of a sanitary landfill. (CO5, CO2)

4. (a) How does bioremediation help in overcoming the problem of soil pollution ? (CO4, CO5, CO3)

(b) Explain the role of Government officials for the disposal of solid wastes. (CO4, CO5, CO3)

(c) 'Exposure of sewage sludge or other types of wet organic waste to high temperature and pressure." Explain such process in detail. (CO4, CO5, CO3)

5. (a) What is the purpose of Up-flow anaerobic sludge blanket reactor ? (CO2, CO1)

(b) Discuss the after effects caused by acid rain with examples. (CO2, CO1)

(c) Discuss the applications of any *five* enzymes which are effective in the cleaning of environmental remnants. (CO2, CO1)

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TBT–405(a)

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

WASTE MANAGEMENT AND UPCYCLING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is waste upcycling ? How does it differ from recycling and down cycling. (CO1, CO2, CO3)

   (b) Give a detailed account waste water treatment ? Explain the process of industrial waste treatment. (CO1, CO2, CO3)

   (c) How does effluent treatment plant help in waste water treatment ? Explain the process in detail. (CO1, CO2, CO3)

2. (a) What are the approaches for obtaining energy from waste ? Explain with suitable examples. (CO3, CO4, CO5)

   (b) How composting can be a sustainable solution for waste upcycling ? Provide any case study in support of your answer. (CO3, CO4, CO5)

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(c) What is particulate matter ? Mention how air pollution control devices help in removal of air pollutants. (CO3, CO4, CO5)

3. (a) What are the hazards of biomedical waste ? Explain biomedical waste management with help of suitable examples. (CO5, CO2)

(b) What is landfill leaching and how does land fill engineering help in landfill management ? (CO5, CO2)

(c) How liquid waste collection and treatment is performed ? Explain in detail. (CO5, CO2)

4. (a) How phytoremediation can be a sustainable solution for tackling the problem of soil pollution ? (CO2, CO4, CO5)

(b) What do you understand by carbon foot printing ? How does it help in tackling environmental pollution ? (CO2, CO4, CO5)

(c) How organic waste can be upcycled into different useful products. Provide suitable examples in support of your answer. (CO2, CO4, CO5)

5. (a) Explain any *two* processes in detail : (CO1, CO2, CO3)

(i) Incineration

(ii) Pyrolysis and Gasification

(iii) Closed loop recycling

(b) What is Landfill ? Why we perform Landfill mining and how does it help in managing the solid waste ? (CO1, CO2, CO3)

(c) What are the challenges and opportunities in exploiting degraded and contaminated land as a sustainable option for bioenergy production ? (CO1, CO2, CO3)

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TBT–405(c)

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER) END SEMESTER EXAMINATION, June, 2023

BIOMATERIALS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Enlist typical metals and ceramics used in human implants with examples. (CO1)

(b) State any *five* differences between natural fiber and synthetic fiber. (CO1)

(c) Explain the following in brief : (CO1)

(i) tensile strength

(ii) bending strength

(iii) bulk modulus

(iv) shear modulus

2. (a) Explain in brief the following terms : bio resorption, biodegradation, bio erosion and bio absorption. (CO2)

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(b) Briefly explain the mechanism of development of biofilm on the surface of implants. (CO2)

(c) What are the factors that effect the biofilm wholly or partially. Explain in brief. (CO2)

3. (a) Briefly explain the following : thermogravimetric analysis and differential scanning calorimetry. (CO3)

(b) Write a short note on the following : (CO3)

(i) acoustic spectroscopy,

(ii) florescence spectroscopy

(iii) ultraviolet visible spectroscopy and

(iv) mass spectroscopy

(c) How does the tissue and immune system response to the newly implanted biomaterial. Explain in brief. (CO3)

4. (a) Write a short note on fabrication of biomaterial via rolling and forming process. (CO4)

(b) Write a short note on injection moulding process with suitable diagram. (CO4)

(c) Explain in brief the drawing process with neat diagram. (CO4)

5. (a) Name and explain in brief the material used for implantation as scaffolds and dental amalgam. (CO5)

(b) What are the advantages of polymer blend over neat polymer. Name any *five* . (CO5)

(c) Describe in brief any *two* processes of preparation of hydrogels. (CO5)

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